

Mechatronic Design and Prototype of a 4-DOFs Hot-Wire CNC Cutting Machine

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Abstract. The subject of this paper is the mechatronic design of a 4-DOFs hot-wire CNC cutting machine, which is mainly composed by a conventional Cartesian robot with 3-DOFs, plus one that corresponds to the rotation of a fork, in order to orient the hot-wire during the cutting. The proposed cutting machine is actuated through four stepper motors and controlled in such a way to obtain a CAD-CAM system, which is able to generate both planar and skew ruled surfaces, because of the 3-dimensional motion of the hot-wire that is installed on the fork (end-effector). A prototype was built and several experimental tests have allowed the validation of the proposed mechatronic design.

Keywords: Mechatronic design · Cartesian robot · CAD-CAM · Hot-wire cutting machine

1 Introduction

This paper describes the main results obtained by the development of a research project, which was supported by INAIL (National Institute for Insurance against Accidents at Work) and aimed to disseminate good practices in the field of the industrial safety by devoting particular attention to the Navigation District of the Italian region of Lazio. Moreover, some design solutions were proposed to avoid ergonomic risks and to prevent accidents, health injuries and occupational diseases to the operators in that specific industrial field. In fact, the manufacturing process of yachts and fiberglass boats requires the use of polyurethane foam beams, which are installed inside the female mold of the boat, in order to generate an internal structure of stringers and longerons, which are then covered by a tissue of composite material to give a final resistant structure. These stringers and longerons are characterized by different shapes because of the hydrodynamic design specifications of the boat, which can be approximated by cylindrical ruled surfaces, from which the common use of manual band saws.

Consequently, this step of the manufacturing process is very dangerous for the workers since the risk of accidents is very high during the cutting of the polyurethane foam blocks and again, the ergonomic conditions during the manual positioning of each piece of stringers and longerons are also very bad, because the workers have to spend long time down, inside the female mold of the boat.

Thus, according to these scientific motivations and among different solutions, the mechatronic design of a 4-DOFs hot-wire CNC foam cutting machine was developed, built and experimentally tested at LAPI (Laboratory of Industrial Design) of the University of Cassino and Southern Lazio.

This choice was done by taking into account that common robots of the industrial environment are usually designed and built with the aim to perform different tasks, such as welding, painting, assembly, pick and place, packaging, labelling and so on. The fundamentals for analyzing different kinematic chains and robotic devices can be found in [1] for both industrial and non-industrial robotics, as in the case of mobile and walking robots [2, 3]. Moreover, Cartesian robots [4–7], sometimes called gantry robots, are mechatronic devices that use motors and linear actuators to position a gripper or a tool, with the aim to grasp and release an object or to perform a specific working task. They make linear movements in three axes, which are mutually orthogonal, as for a 3D Cartesian reference frame. The typical pick and place operation is performed through a suitable gripper, which can be designed according to the specific application, as in reported in [8]. A force control of a 2-axes robotic system was proposed in [9] with the aim to improve the hotwire cutting process of polystyrene plates, even if, in many practical applications of the industrial environment this high accuracy is not required. Thus, the proposed machine can perform the hot-wire cutting process of polystyrene, but not polyurethane foam because this will burn instead of melting by creating nasty fumes, which are not allowed in the workplace. The cutting of polyurethane foam can be carried out by using a band saw or an abrasive wire cutter, but the second type is considered more safe than the first. However, the geometric process of cutting is the same when hot or abrasive wire cutters are used, because both cutting techniques give the final shape of the workpiece in the form of ruled surfaces, as cylinders, cones, hyperboloids, which are generated by the motion of a straight line in the space. Of course, the abrasive wire cutter must translate along its axis in order to perform the cutting process by means of abrasion and thus, one more degree of freedom should be foreseen in this type of cutting machine.

Therefore, this paper deals with the mechatronic design of a 4-DOFs hot-wire CNC foam cutting machine, which is mainly composed by a conventional Cartesian robot with 3-DOFs, plus one that corresponds to the rotation of a fork, in order to orient the hot-wire during the cutting. The proposed cutting machine is actuated through four stepper motors and controlled in such a way to obtain a CAD-CAM system, which is able to generate both planar and skew ruled surfaces, because of the 3-dimensional motion of the hot-wire that is installed on the fork (end-effector). A prototype was built and several experimental tests have allowed the validation of the proposed mechatronic design.

2 Mechatronic Design of the Hot-Wire CNC Cutting Machine

The mechatronic design of the 4-DOFs hot-wire CNC cutting machine is proposed. The power block of this automatic machine is mainly composed by a Cartesian robot with a fourth degree of freedom that is associated to a revolute joint installed on the extremity of the third Cartesian axis. The end-effector of the 4-DOFs cutting machine is

represented by a fork to keep the hot wire in tension and to orient it during the cutting process.

Figure 1 shows the power block of the proposed CNC cutting machine, which is composed by the Cartesian robot with three moving axes, X , Y and Z , along with a forth d.o.f. corresponding to the θ rotation angle. Each degree of freedom of the 4-DOFs cutting machine is actuated by a stepper motor and the cutting operation of the workpiece is carried out through the hot-wire in the XZ plane and along with its orientation because of the θ rotation. Thus, the displacement along the Y -axis corresponds to the forward motion in the case of workpiece fixed at the base.

Referring to Fig. 2, the control block of the proposed CNC cutting machine is composed by (1) PC to program and to transmit the data; (2) the electronic board to control the four stepper motors by means of the corresponding electric drives M1, M2, M3 and M4. The CAD/CAM process is sketched in the flow-chart of Fig. 3.

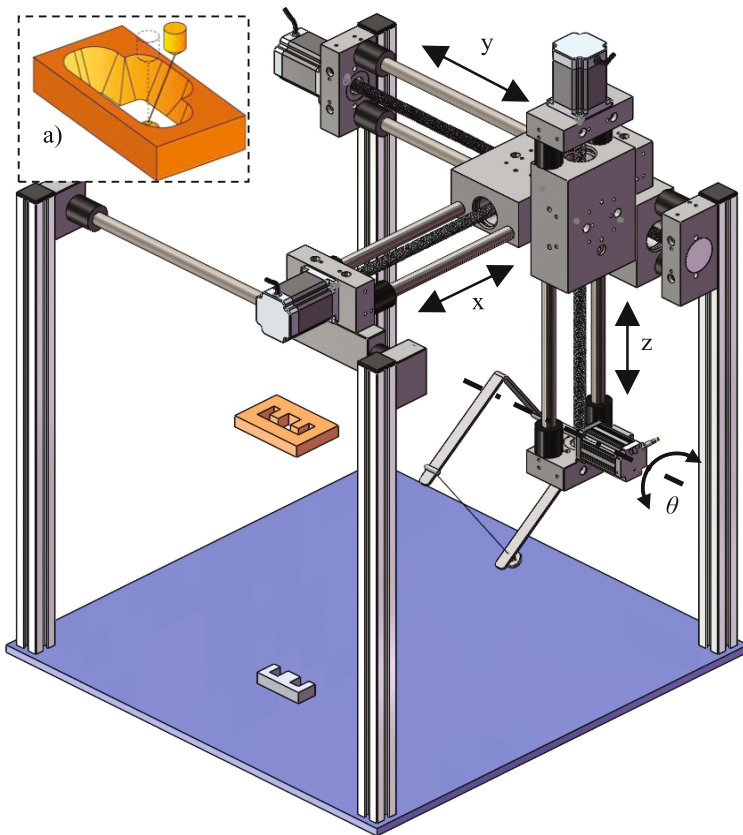


Fig. 1. Mechanical design of the proposed CNC hot-wire cutting machine.

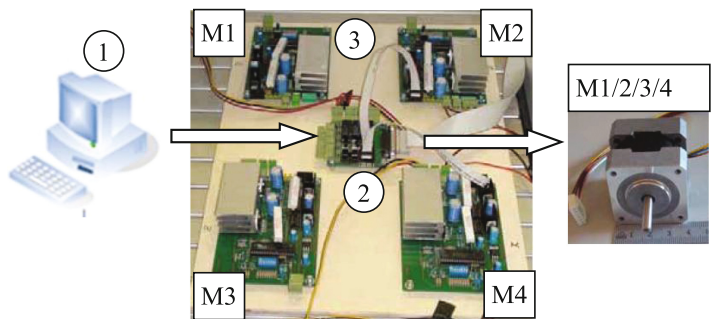


Fig. 2. Scheme of the electronic connections; (1) PC; (2) electronic board of type CN4ACE for controlling 4 axes; (3) four equal electronic boards of type CNCARM04 for controlling the stepper motors.

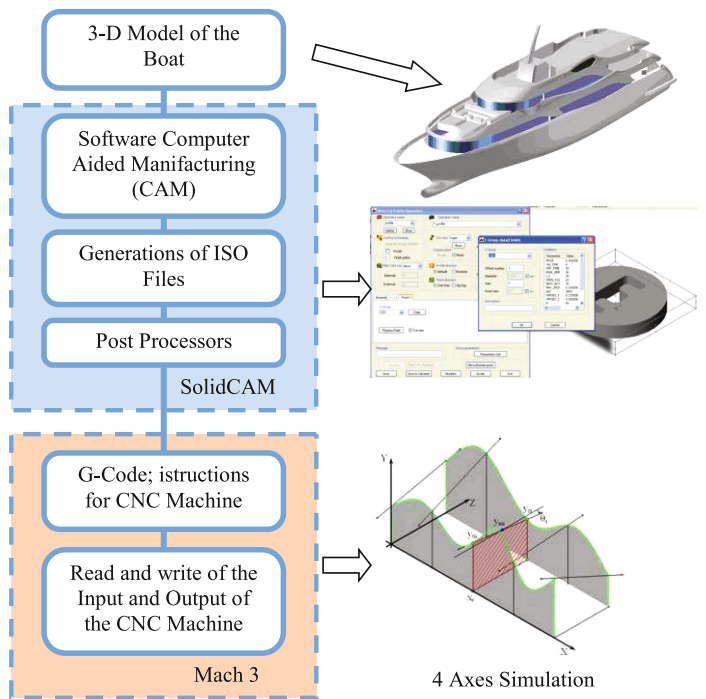


Fig. 3. Sketch of the CAD/CAM process.

3 Surface Geometry of the Boat Hull

In the CAD design for the marine industry the main curves and surfaces used are N.U.R.B.S., and these will be considered in the following. Curves and surfaces N.U.R.B.S. (Non Uniform Rational B-Splines) are a generalization of the forms of

Bezier and B-splines and have become the industry standard for the design and surface modeling. They are particularly suitable for the modeling of surfaces with complex curves as easy to handle on the interactive floor and also to the fact that the algorithms created from them are efficient and numerically stable. From an analytical point of view the N.U.R.B.S. are however of polynomial curves and their parameterization is given by the Eq. (1).

$$t \in J = [t_k, t_{n+1}] \subseteq I \subset \mathbb{R} \rightarrow X(t) \in \mathbb{R}^2 \quad X(t) = \frac{\sum_{i=0}^n h_i \cdot B_i \cdot N_i^k(t)}{\sum_{i=0}^n h_i \cdot N_i^k(t)} \quad (1)$$

where n is a fixed natural number, are of $n + 1$ points of $\mathbb{R}^2$ said control points which define the vertices of the control polygon, $N_i^k(t)$ is the i -th B-spline of order k and h_i are non-negative values, said weights, associated with the control points (Fig. 4).

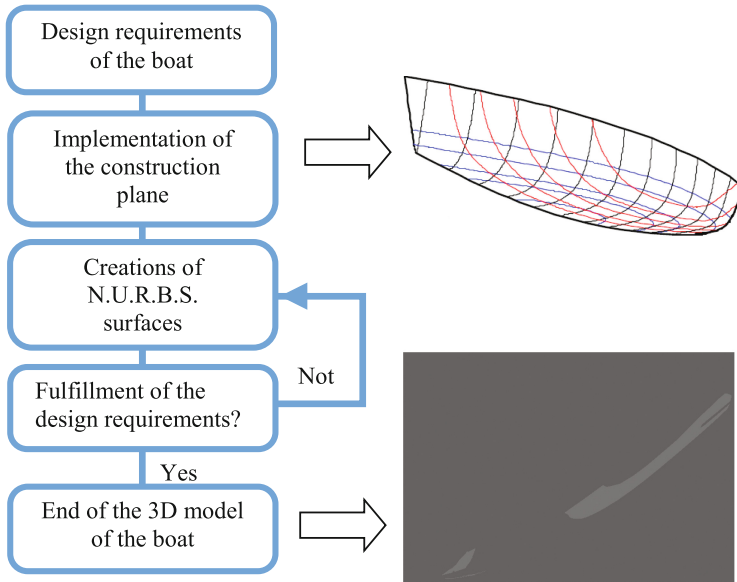


Fig. 4. Functional scheme to obtain the shape of the boat.

4 Laboratory Test Bed and Experimental Tests

According to the proposed mechatronic design, a prototype of the CNC Foam Cutting Machine was built and tested by means of the experimental test-bed of Fig. 5, which shows:

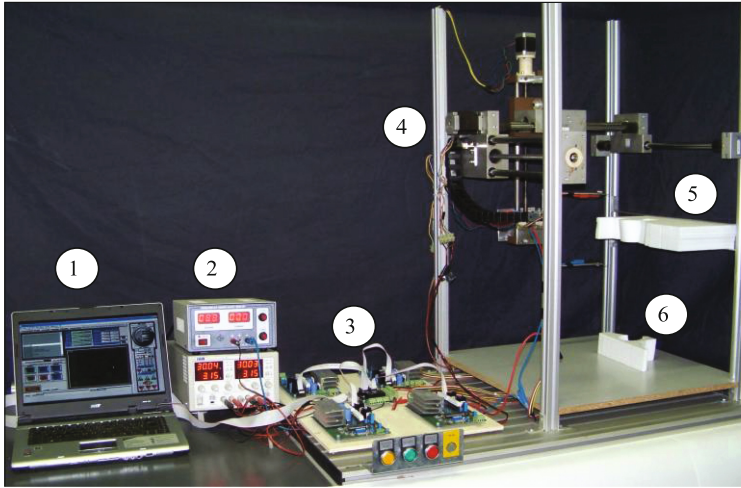


Fig. 5. The built test-bed: (1) PC; (2) power supply; (3) electronic board for controlling the four stepper motors; (4) CNC machine; (5) workpiece during the cutting process; (6) cut out piece.

1. a PC that is supplied with an USB card of type DB25 25-Pin Parallel Port Cable Adapter, which is connected to the electronic board CN4ACE for controlling four different stepper motors independently;
2. two different power supplies, one of 30 VDC for the electronic board and one for the regulations of the temperature of the hot wire cutting;
3. four equal electronic boards of type CNAM04 for controlling each motor;
4. the CNC machine that is composed by 3 linear axes and one rotational axis for the generation of the ruled surface (the maximum displacements for the X, Y and Z axes are 200, 250 and 400 mm, while the end-effector is able to turn of 360°);
5. Styrofoam piece for the shear tests;
6. Cut out styrofoam piece.

The chosen screw-and-nut system is composed by a screw and a cylindrical nut screw with trapezoidal profile. The nominal diameter of the screw is 12 mm with a step of 3 mm, this means that using motors of 1.8° (pitch angle), the resolution for the three axes can be expressed as:

$$\Delta X_{\min} = \Delta Y_{\min} = \Delta Z_{\min} = \frac{p}{360^\circ} \cdot \theta = \frac{3}{360^\circ} \cdot 1.8^\circ = 0,015 \text{ [mm]} \quad (2)$$

Likewise, by using a speed reducer with a transmission ratio of 720:1, the resolution angle for each step of the 4th axis can be obtained by:

$$\Delta \theta = \frac{360^\circ}{200 \cdot 720} = 0.0025^\circ \quad (3)$$

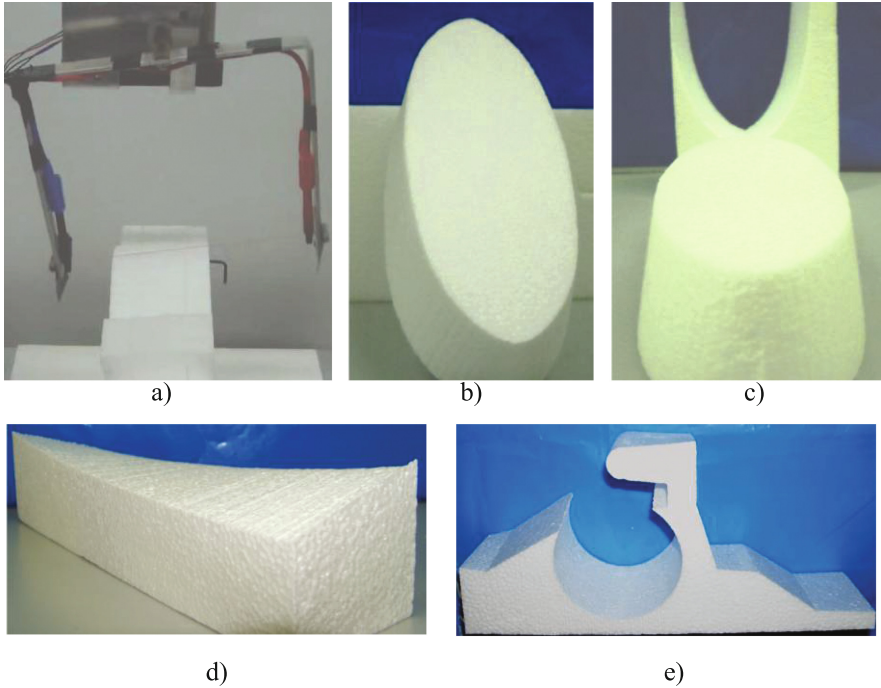


Fig. 6. Few examples of the cut out styrofoam pieces: (a) zoom-in of cutting step; (b) truncated elliptical cone; (c) truncated circular cone; (d) longitudinal styrofoam piece with the variations of the angle from -15° to 15° ; (e) complex styrofoam piece with the variations of the angle from -20° to 15° .

Figure 6a shows the hot-wire cutting task of a skew ruled surface, which can be generated by the rotation of the end-effector fork according to its forward motion.

In Fig. 6b, it is shown a truncated elliptical cone. The lateral inclination of the cut is 10° . In Fig. 6c, it is shown a circular truncated cone, also in this case, the lateral inclination of the cut is 10° . In Fig. 6d, it is shown longitudinal Styrofoam piece with the variations of the angle from -15° to 15° , thanks to this example, it can be shown that this CNC Foam Cutting Machine is able to realize the machining of the ruled surfaces. Finally, in Fig. 6e, it is shown an example of complex styrofoam piece with the variations of the angle from -20° to 15° .

High accuracy is usually not required for the proposed hot-wire cutting process or in the case of abrasive wire cutters, because of the specific application.

5 Conclusions

This paper describes the main results obtained by the development of a research project, which was supported by INAIL (National Institute for Insurance against Accidents at Work) and aimed to disseminate good practices in the field of the

industrial safety by devoting particular attention to the Navigation District of the Italian region of Lazio.

In particular, the mechatronic design of a 4-DOFs hot-wire CNC foam cutting machine has been proposed with the aim to obtain a CAD-CAM system, which is able to manufacture any piece of polystyrene with a shape of ruled surfaces.

This automatic machine is mainly composed by a conventional Cartesian robot with 3-DOFs, plus one that corresponds to the rotation of a fork, in order to orient the hot-wire during the cutting process, which allows the manufacturing of skew ruled surfaces and not only curvilinear cylindrical surfaces.

A prototype was built at LAPI (Laboratory of Industrial Design) of the University of Cassino and Southern Lazio and several experimental tests have been carried out with the aim to validate the proposed CAD-CAM system, which allows the manufacturing of different workpiece having a shape composed by ruled surfaces. The corresponding CAD model is the only input required.

References

1. Angeles J (2014) *Fundamentals of Mechanical Robotic Systems: Theory, Methods, and Algorithms*, 4th edn. Springer, Cham
2. Gonzalez Rodriguez A, Gonzalez Rodriguez A, Rea P (2011) A new articulated leg for mobile robots. *J Ind Robot* 38(5):521–532
3. Figliolini G, Ceccarelli M (2001) Climbing stairs with EP-WAR2 biped robot. In: *CD Proceedings of the IEEE international conference on robotics and automation (ICRA 2001)*, Seoul, Korea, pp 4116–4121
4. Sanchez-Sanchez P, Reyes-Cortés F (2010) Cartesian control for robot manipulators. In: Jimenez A, Al Hadithi BM (eds) *Robot manipulators, trends and development*. InTech, Rijeka, pp 165–211
5. Korayem MH, Khanali T (2002) Design and manufacturing of prismatic robot for assembly line. In: *Proceedings of the first international conference on information technology & applications*, Bathurst, Australia, paper no. 114–9
6. Brunnenkreef JEJ (2007) Modeling and control of a four degree of freedom Cartesian robot for the packaging industry. Master Thesis, University of Twente, Enschede, Netherlands
7. Altintas A (2010) New approach to 3-axis cylindrical and Cartesian type robot manipulators in mechatronics education. *Electron Electr Eng* 106(10):151–154
8. Ceccarelli M, Figliolini G, Ottaviano E, Mata AS, Criado EJ (2000) Designing a robotic gripper for harvesting of horticulture products. *Robotica* 18(1):105–111
9. Gallina P, Mosca R, Pascutto P (2005) Optimized hotwire cutting robotic system for expandable polystyrene foam. In: *Advanced manufacturing systems and technology*. CISM international centre for mechanical sciences, vol 486, pp 377–386